Arshian Ali

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FAST NUCES ISLAMABAD Islamabad, Pakistan **GPA: 3.53/4.0**
B.S. in Computer Science *Expected Graduation, May 2028*

SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence, full-stack development, and systems programming. Currently a Full-Stack Development Intern at **Coding The Brains**, building AI-powered web applications. Comfortable across the stack – from training and integrating ML models to designing databases, APIs, and production-facing UIs.

TECHNICAL KNOWLEDGE

- **Languages:** Python, C++, JavaScript/TypeScript, SQL
- **AI & ML:** Neural networks & backpropagation, search algorithms (BFS/DFS/A*), genetic algorithms, classification & clustering, computer vision
- **Web & Backend:** React, Vite, Node.js, FastAPI, REST APIs, PostgreSQL, MySQL
- **Systems:** Process scheduling, synchronization, shared memory & IPC, x86 Assembly
- **Tools:** Git, Linux, NumPy/Pandas, VS Code

EXPERIENCE

Coding The Brains Wah, Pakistan
Full-Stack Development Intern 2026 – Present

- Build and ship AI-powered full-stack web applications, integrating ML models into production-facing products
- Developed **MySpeakScore**, an AI-driven speech evaluation tool that automatically scores speech and generates feedback
- Developed **Fitness Buddy** (React/Vite, Node.js, PostgreSQL), a fitness platform for coaches and clients with workout/macro tracking and an AI chat widget for personalized guidance
- Contributed to computer vision pipelines for image and video-based detection

PROJECTS

Stix – Local-first Markdown workspace *React, IndexedDB, Fuse.js, CryptoJS*

- Cloudless Markdown editor: all documents, media, and search run entirely on-device, with no backend or accounts
- Built a version control system (snapshots, branching, diffing) and an encrypted, password-protected `.stix` backup format

PitWall Pro – F1 race strategy simulator *Python, FastAPI, MySQL*

- Strategy engine built on real F1 telemetry, predicting tyre degradation and optimal pit windows
- Used heuristic optimization to compare 1-stop vs. 2-stop strategies, with live telemetry visualizations

SimpleML – Neural network library, C++, no dependencies

- Implemented tensors, backpropagation, and optimizers from first principles to understand deep learning internals

Chrono Rift & Mario Assembly – Systems projects in C++ and x86 Assembly

- A terminal combat game using shared memory and multithreading to synchronize a real-time AI opponent; and a full Super Mario Bros recreation built from scratch in x86 Assembly

ACHIEVEMENTS & ACTIVITIES

- Competitive Programming – FPSC and ICPC contests at GIKI
- Datum Labs Data Visualization Competition – analytical storytelling under time constraints
- Model United Nations & Debate – 4 years of diplomacy, negotiation, and public speaking